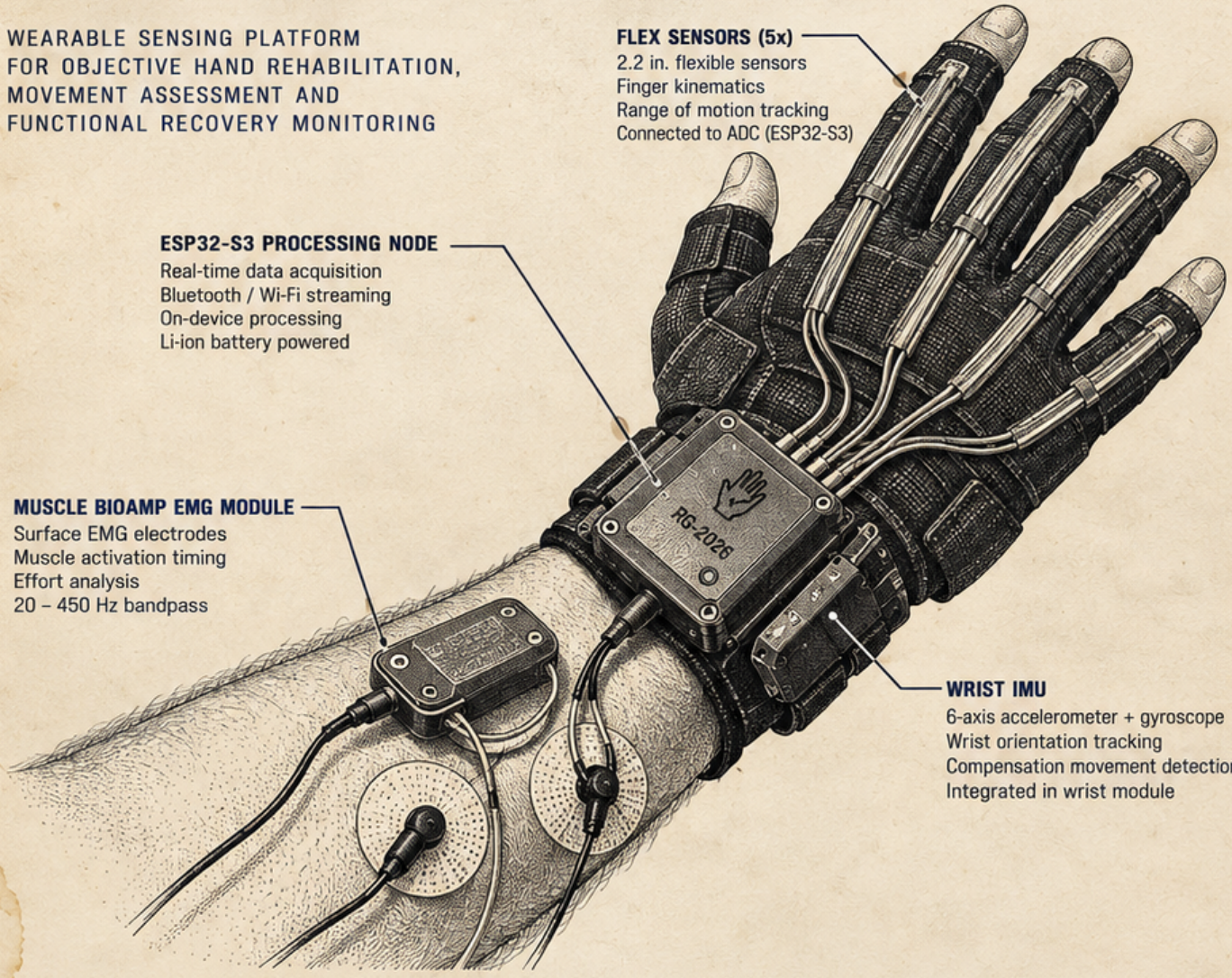
MULTIMODAL HAND KINEMATIC & ELECTROMYOGRAPHIC ASSESSMENT SYSTEM

MODEL RG-2026 · OPERATOR'S MANUAL & CLINICAL GUIDE

WEARABLE SENSING PLATFORM
FOR OBJECTIVE HAND REHABILITATION,
MOVEMENT ASSESSMENT AND
FUNCTIONAL RECOVERY MONITORING



FLEX SENSORS (5x)
2.2 in. flexible sensors
Finger kinematics
Range of motion tracking
Connected to ADC (ESP32-S3)

ESP32-S3 PROCESSING NODE
Real-time data acquisition
Bluetooth / Wi-Fi streaming
On-device processing
Li-ion battery powered

MUSCLE BIOAMP EMG MODULE
Surface EMG electrodes
Muscle activation timing
Effort analysis
20 – 450 Hz bandpass

WRIST IMU
6-axis accelerometer + gyroscope
Wrist orientation tracking
Compensation movement detection
Integrated in wrist module

HAND REHABILITATION MEASUREMENT PLATFORM — CLINICAL OPERATOR GUIDE

FLEX KINEMATICS



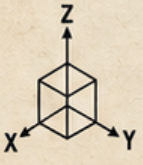
- 5 x 2.2 in. flexible sensors
- ADC-calibrated angles
- Finger ROM & timing
- Movement pattern analysis

SURFACE EMG



- Muscle BioAmp signal chain
- 20 – 450 Hz bandpass
- Muscle activation timing
- Effort and coordination

6-AXIS IMU



- Wrist orientation tracking
- Compensation detection
- Movement stability index
- Integrated with main module

⚠ IMPORTANT SAFETY & REHABILITATION INSTRUCTIONS

- ☐ For clinical and research use only.
- ☐ Do not use on open wounds or damaged skin.
- ☐ Ensure proper electrode placement and skin preparation.
- ☐ Do not expose the system to excessive moisture.
- ☐ Do not pull sensor cables or apply excessive force.
- ☐ Use only approved chargers and batteries.
- ☐ Discontinue use if discomfort or skin irritation occurs.
- ☐ Follow clinical supervision protocols at all times.

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